

Chapter 30. Mechanical Ventilation for Severe Asthma

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Mechanical Ventilation for Severe Asthma: Introduction

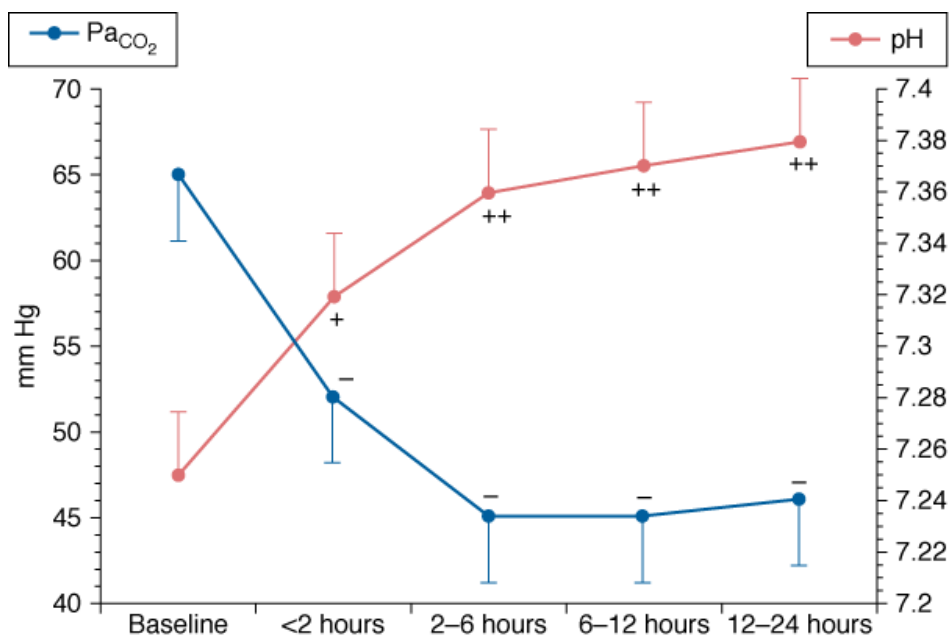
It is estimated that 6000 to 10,000 patients require mechanical ventilation for acute asthma in the United States each year.¹ Asthma exacerbations that lead to mechanical ventilation typically progress over 1 or more days, with viral infections being the most common etiology (Table 30-1).² Airways typically show extensive mucous plugging and eosinophilic infiltration with edema, explaining the often limited immediate response to bronchodilators.³ Approximately 20% of patients experience a more explosive onset, with attacks occurring over minutes to hours.²⁻⁴ The precise cause of “sudden asphyxial” asthma is not always apparent, but recognized triggers include aeroallergens, nonsteroidal antiinflammatory agents, airway irritants, and emotional distress.^{2,4} Rapid-onset attacks are characterized by profound bronchospasm with minimal mucous plugging, explaining both their sudden onset and rapid resolution.³

Table 30-1: Patterns of Near-Fatal Asthma

	Slow-Onset	Rapid-Onset
Time course (onset)	One or more days	Minutes to hours
Triggers	Virus, unknown	Aeroallergen, nonsteroid antiinflammatory drug, airway irritant, emotional stress, unknown
Frequency	~80%	~20%
Mechanisms of airflow obstruction	Mucous plugging and airway edema > bronchospasm	Bronchospasm (“dry airways”)
Airway inflammation	Eosinophils	Neutrophils
Response to treatment	Slow: minimal response to initial bronchodilators	Rapid: good response to initial bronchodilators
Prevention	Steroids early in exacerbation	Avoid triggers
Duration intubation	Often several days	Often < 24 hours

Regardless of the mode of onset, life-threatening asthma is associated with a markedly increased airway resistance, pulmonary hyperinflation, and high physiologic dead space, which together lead to hypercapnia and risk of respiratory arrest.⁵ Hypercapnia per se is not an indication for intubation, as most asthma exacerbations with hypercapnia do respond to bronchodilator therapy and others may be successfully managed with noninvasive ventilation (Fig. 30-1).⁶⁻¹¹ Inhalation of a helium–oxygen gas mixture (heliox) may also reduce work of breathing and might decrease the likelihood of intubation.^{11,12}

Figure 30-1



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Change in partial pressure of arterial carbon dioxide (Pa_{CO_2}) and pH associated with the use of noninvasive positive-pressure ventilation by face mask in patients with severe asthma. (Used, with permission, from Meduri GU, Cook TR, Turner RE, et al. Noninvasive positive pressure ventilation in status asthmaticus. *Chest*. 1996;110:767-774.)

Indications for intubation include respiratory arrest, depressed level of consciousness, or progressive fatigue and exhaustion. Rapid-sequence intubation has been advocated by some experts,¹³ but a high degree of confidence in being able to intubate is a prerequisite before use of a paralytic because it may be difficult to achieve effective bag-mask ventilation in patients with markedly increased airway resistance. Regardless of the technique used, intubation should be performed by the most skilled operator present. Prolonged airway manipulation and repeated failed intubations in the setting of fulminant asthma may prove catastrophic.

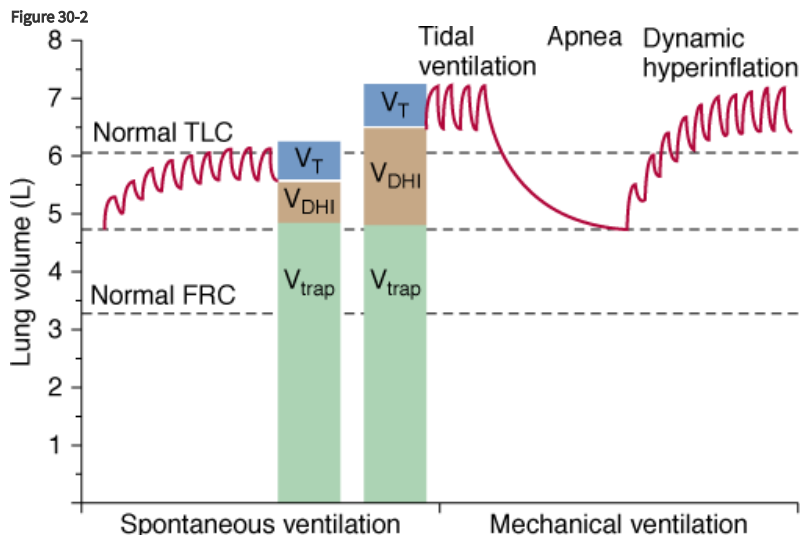
Various drugs have been used to facilitate intubation. Ketamine does not cause respiratory depression or hypotension, but may increase laryngeal reflexes and predispose to laryngospasm with excessive upper airway manipulation.¹⁴ Propofol decreases airways resistance after intubation as compared with etomidate or thiopental.¹⁵ Etomidate, however, causes less hypotension than propofol and may be a better alternative in hemodynamically unstable patients.¹⁴ Indeed, many of the drugs used to facilitate intubation reduce vascular tone, which, together with a sudden decrease in venous return secondary to pulmonary hyperinflation, may result in postintubation hypotension.¹⁶ The latter should be anticipated and managed by rapid fluid administration and manual ventilation via an Ambu bag at a slow rate to limit the severity of hyperinflation.¹⁷ Postintubation hypotension that is profound and refractory to these first-line interventions may be managed with a bolus dose of a rapidly acting vasopressor, such as [phenylephrine](#), vasopressin, or [epinephrine](#).

Ventilator Management

Controlled hypoventilation with permissive hypercapnia was first proposed as a ventilator strategy for severe asthma almost 30 years ago.¹⁸ In brief, the rationale for this approach is that hypercapnia poses less risk than does markedly increased lung volume.^{19,20} Although randomized studies of controlled hypoventilation in asthma are lacking, a retrospective analysis suggested that it is associated with better outcomes than a conventional ventilator strategy.¹⁹ To apply controlled hypoventilation rationally, it is essential to understand methods for monitoring pulmonary hyperinflation and how the latter is influenced by the choice of ventilator settings.²¹⁻²⁵ Subsequent discussion of ventilator strategy and monitoring will be limited to use of volume-cycled ventilation, the preferred method for managing patients with severe airflow obstruction.²³

Pulmonary Hyperinflation: Mechanism and Assessment

Patients with acute severe asthma breathe near their total lung capacity; with mechanical ventilation, lung volumes may increase further (Fig. 30-2).²³ During mechanical ventilation, dynamic hyperinflation is initiated when there is insufficient time during expiration for complete exhalation of delivered tidal volume, resulting in an increase in end-expiratory lung volume and auto-positive end-expiratory alveolar pressure (auto-PEEP). With subsequent breaths, a progressive increase in lung volume leads to improved expiratory gas flow because of higher elastic recoil pressure and an increase in airways diameter, permitting the entire tidal volume to be exhaled (Fig. 30-3). Although dynamic hyperinflation may be an adaptive process that enhances expiratory flow, it can result in severe hypotension and barotrauma, life-threatening complications that are a consequence of extreme alveolar overdistension.

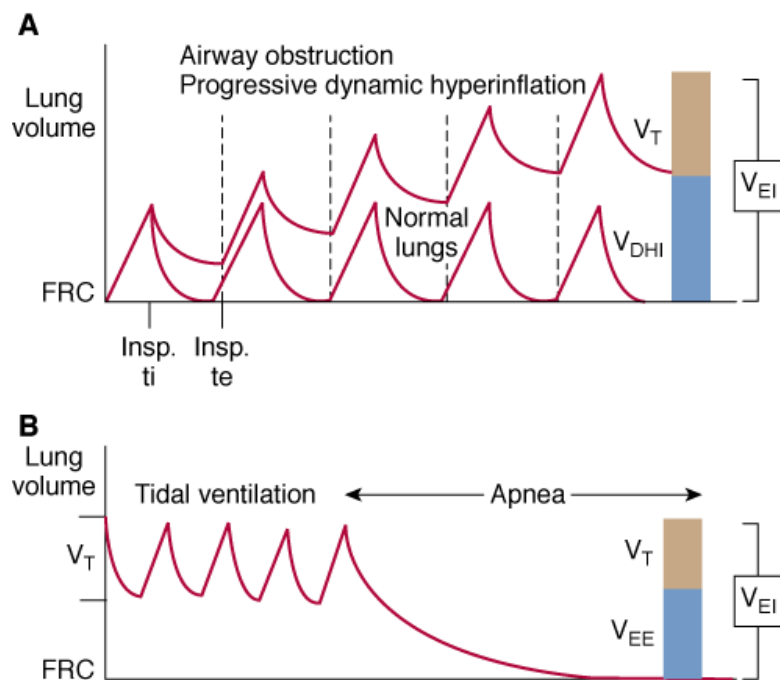


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Lung volumes in severe asthma during spontaneous ventilation and with mechanical ventilation. Tidal breathing occurs near total lung capacity (TLC) during spontaneous ventilation and above TLC during mechanical ventilation. After delivery of tidal volume (V_T) during mechanical ventilation, a prolonged apnea permits exhalation of volume produced by dynamic hyperinflation (V_{DHI}); lung volume at the end of apnea, however, remains well above normal functional residual capacity (FRC) because of gas trapped behind occluded airways (V_{trap}). (Adapted, with permission, from Tuxen et al.²³)

Figure 30-3



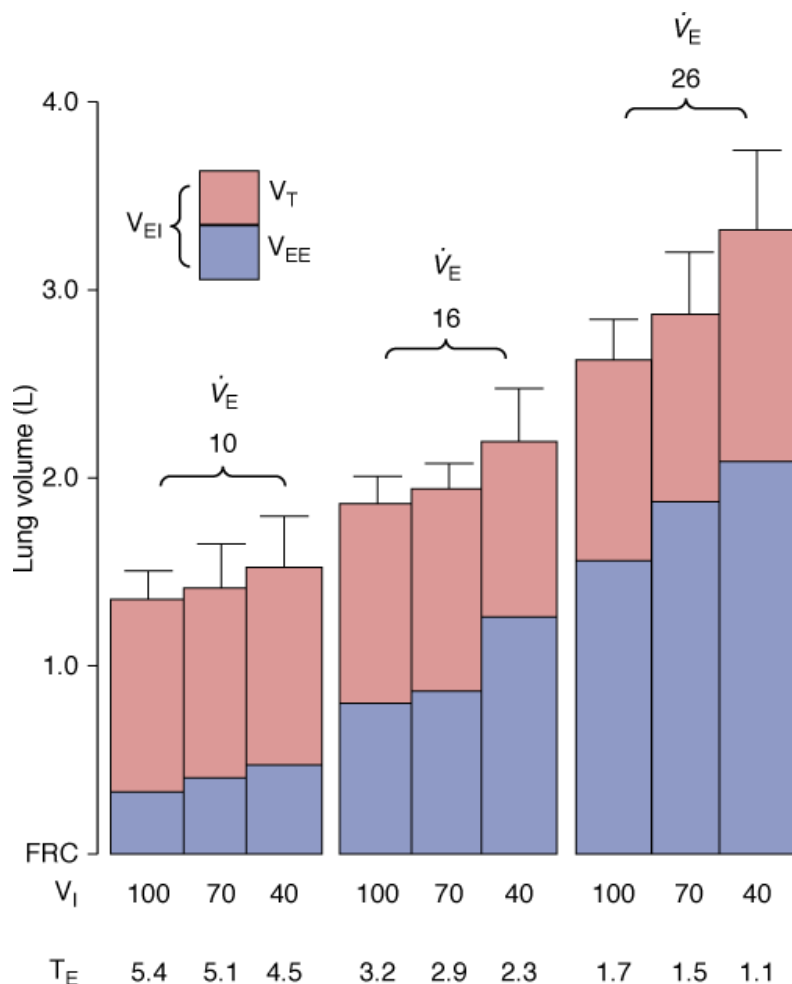
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A. Dynamic hyperinflation. FRC , functional residual capacity; V_{DH} , volume (above FRC) caused by dynamic hyperinflation; V_{EI} , volume (above FRC) at end inspiration; V_T , tidal volume. (Adapted, with permission, from Tuxen. *Am J Respir Crit Care Med*. 1994;150:1722–1737.) **B.** Measurement of V_{EI} and V_{EE} (volume above FRC at end expiration) by use of a prolonged apnea. V_{EE} (volume at end expiration) and V_{DH} (volume produced by dynamic hyperinflation) are equivalent. (Used, with permission, from Tuxen DV, Lane S. The effects of ventilatory pattern on hyperinflation, airway pressures, and circulation in mechanical ventilation of patients with severe air-flow obstruction. *Am Rev Respir Dis*. 1987;136:872–879.)

The severity of hyperinflation in severe asthma varies among patients and within individual patients over time. Serial monitoring of dynamic hyperinflation is important for identifying an increased risk of hypotension and barotrauma and to assess changes in the severity of airflow obstruction in response to therapy.

Two methods of monitoring the severity of airflow obstruction in status asthmaticus have been used: measurement of total exhaled volume during a prolonged apnea beginning at end inspiration and assessment of airway pressure. Tuxen et al^{21–24} used a prolonged apnea to assess dynamic hyperinflation in patients with severe asthma, with the total amount of gas exhaled during the apnea defined as the volume (above functional residual capacity) at end inspiration (V_{EI}). The volume at end expiration (V_{EE}) represents the increase in lung volume caused by dynamic hyperinflation and is calculated by subtracting tidal volume from V_{EI} (see Figs. 30-3 and 30-4). In one study, hypotension and barotrauma were more reliably predicted by V_{EI} than by airway pressures.²⁴ Nonetheless, measurement of V_{EI} and V_{EE} has not gained widespread clinical acceptance, in part because of the need for paralysis.²³

Figure 30-4



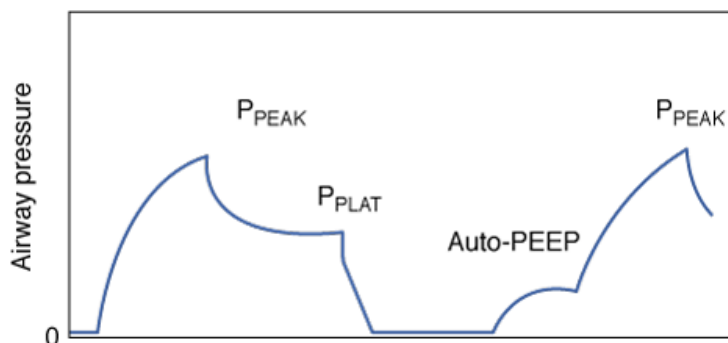
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Effect of minute ventilation ($\dot{V}_E$) and inspiratory flow rate (V_I) on dynamic hyperinflation during mechanical ventilation for severe asthma.

V_{EE} , volume at end expiration; V_{EI} , volume at end inspiration; V_T , tidal volume. (Used, with permission, from Tuxen DV, Lane S. The effects of ventilatory pattern on hyperinflation, airway pressures, and circulation in mechanical ventilation of patients with severe air-flow obstruction. *Am Rev Respir Dis*. 1987;136:872–879.)

A more commonly used monitoring method is assessment of airway pressures (Fig. 30-5). As described initially,¹⁸ controlled hypoventilation in status asthmaticus targeted a peak airway pressure (P_{PEAK}) below 50 cm H₂O, a strategy used by others subsequently.²⁶ P_{PEAK} , however, is highly dependent on inspiratory flow-resistive properties and may not accurately reflect the degree of hyperinflation.^{23,27} Patients with asthma often have a P_{PEAK} above 50 cm H₂O when managed with high inspiratory flow rate (see below), but their risk of barotrauma may be negligible.²⁷ Furthermore, P_{PEAK} may not consistently reflect the reduction in dynamic hyperinflation after expiratory time is prolonged, presumably because of an increase in airways resistance as lung volume decreases (Fig. 30-6).²⁵

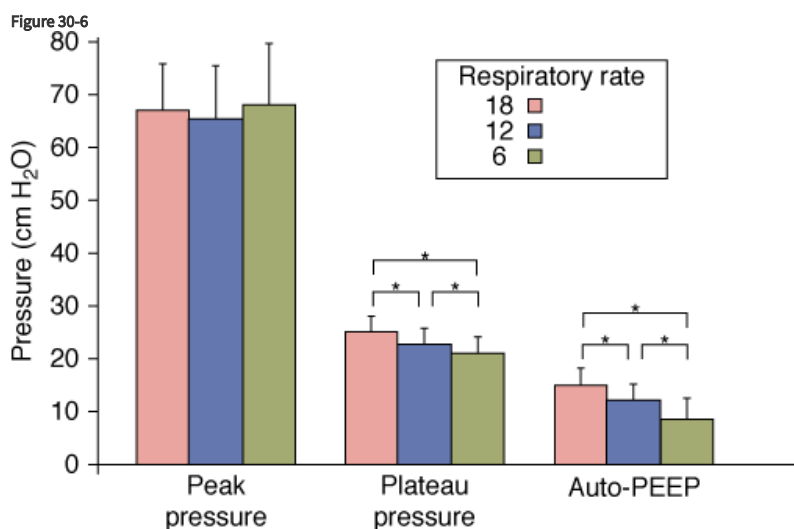
Figure 30-5



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Proximal airway pressure during an end-inspiratory and end-expiratory airway occlusion. End-inspiratory occlusion is followed by an initial rapid fall in airway pressure, secondary to intrinsic airways resistance and then a more gradual fall in pressure secondary to gas redistribution and tissue resistance. P_{PEAK} , peak airway pressure; P_{PLAT} , plateau airway pressure; auto-PEEP, auto-positive end-expiratory pressure.



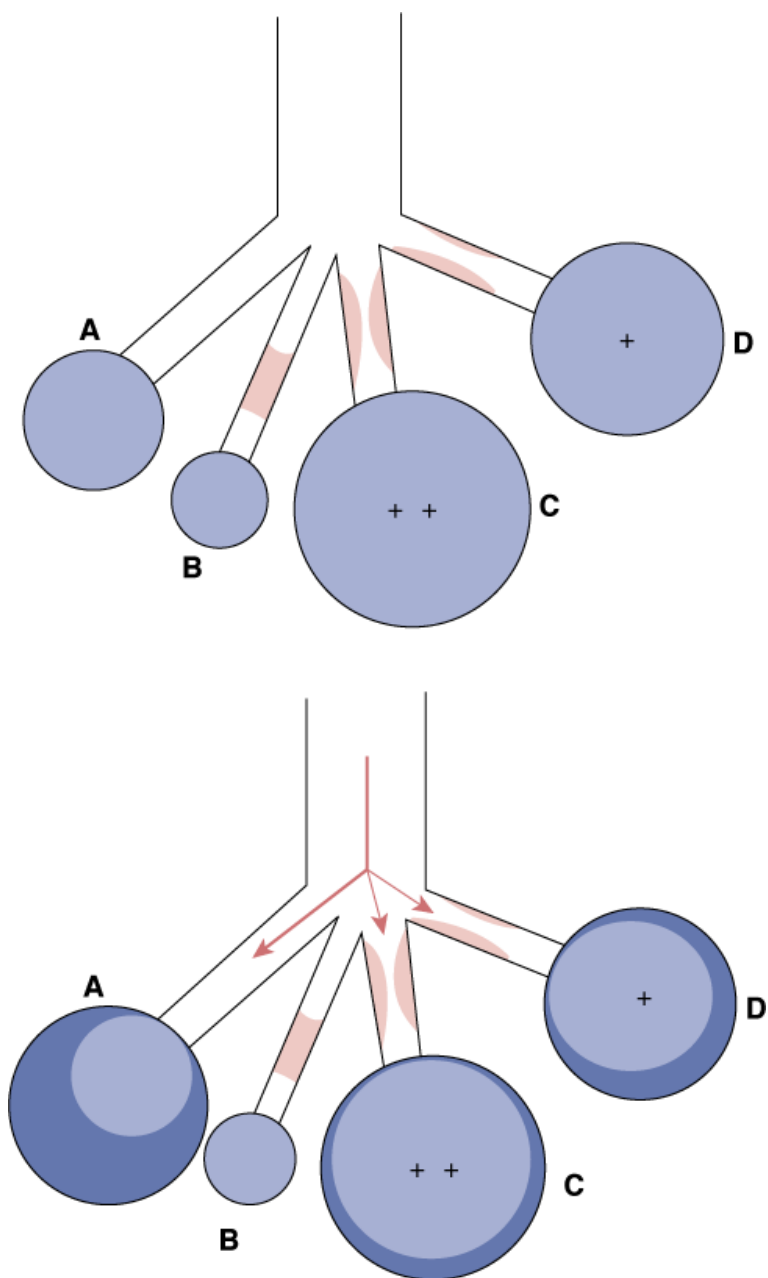
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Peak airway pressure, plateau pressure, and auto-PEEP at respiratory rates of 18, 12, and 6 breaths/min during mechanical ventilation for severe asthma ($n = 12$). (Used, with permission, from Leatherman JW, McArthur C, Shapiro RS. Effect of prolongation of expiratory time on dynamic hyperinflation in mechanically ventilated patients with severe asthma. *Crit Care Med*. 2004;32:1542–1545.)

Plateau airway pressure (P_{PLAT}) may be a better parameter for monitoring lung hyperinflation in status asthmaticus (see Fig. 30-5).^{23,27–31} Patients with severe airflow obstruction typically have near-normal respiratory system compliance and an increase in P_{PLAT} results from dynamic hyperinflation. At constant tidal volume, changes in the degree of dynamic hyperinflation in response to bronchodilators or manipulation of expiratory time can be inferred from changes in P_{PLAT} .²⁵ Because P_{PLAT} provides an estimate of maximal alveolar pressure, it might (in theory) help to predict risk of alveolar rupture. It should be noted, however, that P_{PLAT} represents the *average* end-inspiratory alveolar pressure and the maximal alveolar pressure in some units will be higher (Fig. 30-7).³¹ The safe threshold for P_{PLAT} is not well defined in status asthmaticus, but an acceptable upper limit of 25 to 30 cm H₂O has been suggested most often (Table 30-2).^{23,27,30,31}

Figure 30-7



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Top. Effect of varying degrees of airway obstruction on end-expiratory alveolar volumes and pressures. *Bottom.* Expected distribution of the tidal volume on application of pressure during mechanical ventilation in the setting of inhomogeneous obstruction (With kind permission from Springer Science and Business Media: Oddo et al. Management of mechanical ventilation in acute severe asthma: practical aspects. *Intensive Care Med.* 2006;27:1–10.)

Table 30-2: Mechanical Ventilation for Severe Asthma: One Approach

Ventilator settings (initial)	
Mode	Assist-control
V_T	7 to 9 mL/kg
Rate	10 to 14 breaths/min
V_I	60 to 70 L/min
Waveform	Decelerating or square
PEEP	≤ 5 cm H ₂ O
FI_{O_2}	SaO ₂ > 90%
Sedation/paralysis	
Propofol	2 to 5 mg/kg/h infusion
Fentanyl	50 to 200 μ g/h infusion
Vecuronium	0.1 mg/kg bolus PRN
Therapy of airflow obstruction	
Albuterol-ipratropium MDI	6 puffs qh $\times$ 4, then q1-2h
Methylprednisolone	2 mg/kg/day
Ventilator adjustments	
Goals: $P_{PLAT} < 30$ cm H ₂ O (≤ 25 ideal) and pH ≥ 7.2	
$P_{PLAT} > 30$ cm H₂O $\rightarrow$ decrease $\dot{V}_E$ (rate) - pH < 7.2 and $P_{PLAT} < 25 \rightarrow$ increase $\dot{V}_E$ (rate) - pH < 7.2 and P_{PLAT} 25 to 30 $\rightarrow$ no change (consider buffer if adverse effects of acidosis suspected clinically)	

Abbreviations: MDI, metered-dose inhaler; P_{PLAT} , plateau airway pressure; $\dot{V}_E$, minute ventilation; V_I , inspiratory flow rate; V_T , tidal volume.

When measured by end-expiratory airway occlusion, auto-PEEP provides an estimate of average end-expiratory alveolar pressure. The expected levels of auto-PEEP in status asthmaticus have been infrequently reported, but two studies found average values of 15 and 10 cm H₂O, respectively.^{25,32} In general, changes in auto-PEEP track closely with changes in P_{PLAT} , and either (or both) may be helpful for following the degree of dynamic hyperinflation. On occasion, patients with status asthmaticus who have radiographic evidence of hyperinflation and elevated P_{PLAT} may have unexpectedly low values of measured auto-PEEP when ventilated at very low respiratory rates that encourage airway closure, preventing accurate assessment of end-expiratory alveolar pressure.³³ Auto-PEEP should always be measured during passive expiration, because forceful exhalation often leads to gross overestimation of auto-PEEP and the severity of dynamic hyperinflation, potentially leading to unnecessary restriction of minute ventilation.³⁴

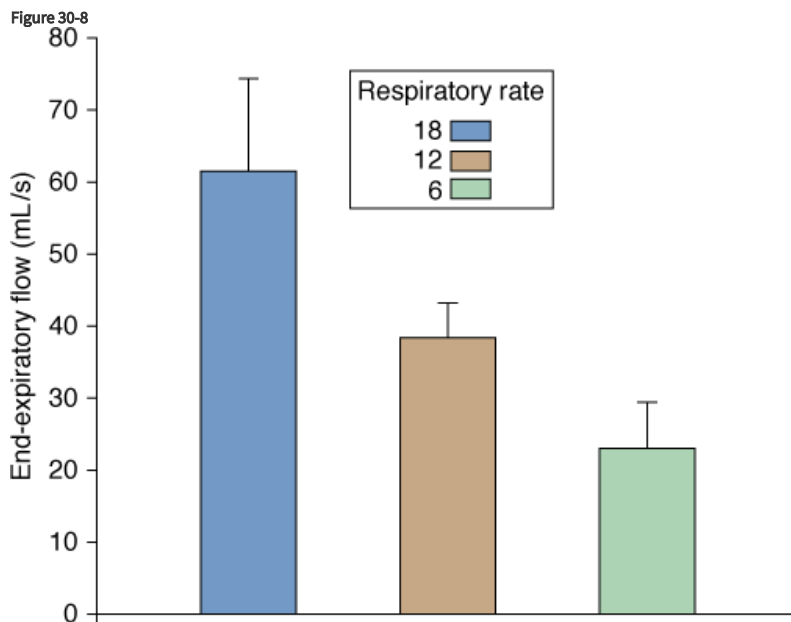
Ventilator Settings

In severe asthma, three key factors determine the degree to which V_{EE} increases during mechanical ventilation: expiratory resistance, tidal volume, and expiratory time. The ventilator settings that most influence the severity of hyperinflation are tidal volume, respiratory rate, and inspiratory flow rate.

Minute Ventilation: Tidal Volume and Respiratory Rate

Tuxen and Lane²¹ examined various ventilator settings in severe asthma and found that minute ventilation was the most important determinant of dynamic hyperinflation (see Fig. 30-4). For a given minute ventilation, the extent of dynamic hyperinflation was similar regardless of the specific combination of respiratory rate and tidal volume.²¹ The degree of hyperinflation became quite marked when minute ventilation was increased from 10 to 16 and 26 L/min; at the highest minute ventilation, both hypotension and barotrauma were noted.²¹

Although these very high levels of minute ventilation clearly pose a risk in the ventilated patient with asthma, the benefit of extreme limitation of minute ventilation, as advocated by some authors,³⁵ is less certain. One study examined the effect of prolongation of expiratory time on dynamic hyperinflation—as assessed by P_{PLAT} and auto-PEEP—when the baseline minute ventilation approximated 10 L/min.²⁵ As expected, P_{PLAT} and auto-PEEP decreased when expiratory time was prolonged, but the magnitude of the reduction was not profound (see Fig. 30-6). For example, when respiratory rate was reduced from 12 to 6 breaths/min (adding 5 seconds to expiratory time), P_{PLAT} and auto-PEEP fell by only 2 to 3 cm H_2O .²⁵ The relatively modest effect on dynamic hyperinflation is understandable given the low expiratory flow rates after a few seconds of expiration (Fig. 30-8). Because flow decreases progressively throughout expiration, the reduction in hyperinflation that results from a given prolongation of expiratory time will depend on the baseline respiratory rate (i.e., less benefit at lower respiratory rates) (Fig. 30-8).



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End-expiratory flow rates at respiratory rates of 18, 12, and 6 breaths/min during mechanical ventilation for severe asthma ($n = 7$). (Used, with permission, from Leatherman JW, McArthur C, Shapiro RS. Effect of prolongation of expiratory time on dynamic hyperinflation in mechanically ventilated patients with severe asthma. *Crit Care Med*. 2004;32:1542–1545.)

Another factor that may limit the degree to which prolongation of expiratory time lessens lung hyperinflation is the presence of gas trapped behind occluded airways. Total pulmonary hyperinflation has two components: a dynamic component that is amenable to ventilator manipulation and a second component caused by gas trapped behind occluded airways, with the latter being unaffected by expiratory time

(see Fig. 30-2). One study suggested that trapped gas accounted for a greater percentage of total pulmonary hyperinflation than did dynamic hyperinflation.²²

In essence, there is often little to be gained by reducing the respiratory rate below 10 to 14 breaths/min when a tidal volume of 7 to 9 mL/kg is used (see Table 30-2). An exception might be when hyperinflation is marked (e.g., P_{PLAT} >30 cm H₂O) or has resulted in complications, because even a small reduction in hyperinflation might have a meaningful clinical impact. Unfortunately, dramatic reductions in hyperinflation may not be easily achieved by ventilator manipulation alone and must await improvement in airflow obstruction.

Inspiratory Flow Rate

It has been suggested that an inspiratory flow rate of 100 L/min and a square waveform be used in patients with status asthmaticus to shorten inspiratory time and lengthen time for expiration.^{23,27,28} This strategy may yield a favorable inspiratory-to-expiratory timing (I:E) ratio, but the impact on dynamic hyperinflation will be modest (Table 30-3). Conventional flow rates (60 to 70 L/min) and a decelerating waveform are appropriate for patients whose minute ventilation has been limited (see Table 30-2).

Table 30-3: Effect of Inspiratory Flow Rate in Severe Asthma

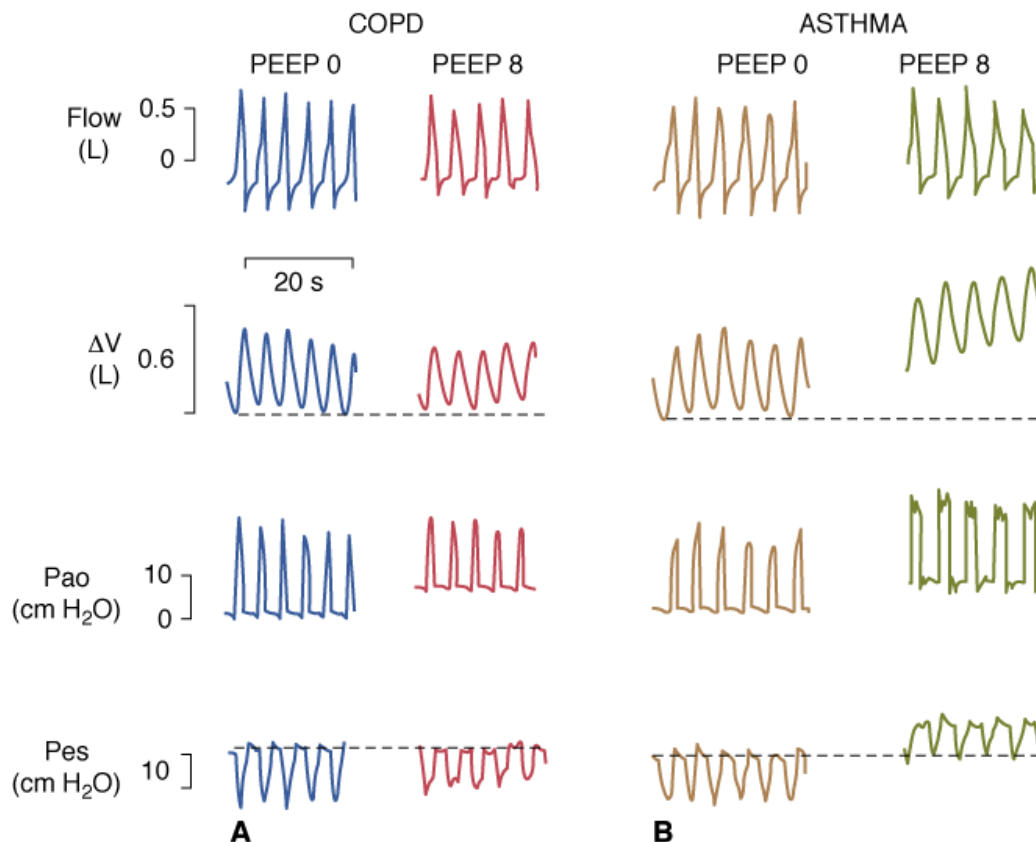
V _I and Waveform	T _I (s)	T _E (s)	I:E Ratio	Δ DHI (ml)	Δ auto-PEEP (cm H ₂ O)
60 L/min, decelerating	1.1	3.2	1:3	—	—
120 L/min, square	0.3	4.0	1:13	~-50	~-1

Note: Assume V_T 600 mL, respiratory rate 14 breaths/min, respiratory compliance 60 mL/cm H₂O, and end-expiratory flow rate approximately 60 mL/s.
Abbreviations: DHI, dynamic hyperinflation; I:E ratio, inspiratory-to-expiratory timing ratio; T_E, expiratory time; T_I, inspiratory time; V_I, inspiratory flow rate.

Applied Positive End-Expiratory Pressure

External PEEP has been used during mechanical ventilation of patients with chronic obstructive pulmonary disease (COPD) to decrease the effort required to trigger the ventilator without increasing lung volume.³⁶⁻³⁸ There is less of a rationale for the use of external PEEP when a patient is undergoing controlled mechanical ventilation. In addition, the impact of applied PEEP on lung volume may be different in asthma and COPD (see Fig. 30-9).^{36,38,39} One study found that external PEEP of 10 to 15 cm H₂O increased the lung volumes of ventilated patients with severe asthma, suggesting that minimal PEEP (≤5 cm H₂O) levels should be used in this setting³⁹ (see Table 30-2). A recent study claimed that the response to PEEP in patients with airflow obstruction is variable, with the effect being an increase, decrease, or no change in lung volume.⁴⁰ The likelihood of a beneficial effect of external PEEP in asthma would seem to be minimal, and if a trial of PEEP is attempted, it should be abandoned immediately if P_{PLAT} increases.

Figure 30-9

(Auto-PEEP = 10 cm H₂O)

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Response to 8 cm H₂O of PEEP in COPD and asthma during proportional-assist ventilation. Esophageal pressure (*Pes*) indicates a similar reduction in inspiratory effort in both patients. External PEEP did not affect lung volume (*V*) or airway opening pressure (*Pao*) in COPD, but both were increased by PEEP in asthma. (Adapted, with permission, from Ranieri VM, Grasso S, Fiore T, Giuliani R. Auto-positive end-expiratory pressure and dynamic hyperinflation. *Clin Chest Med*. 1996;17(3):379–394.)

Gas Exchange: Hypercapnia

Hypercapnia is common during mechanical ventilation of patients with severe asthma, with an average partial pressure of arterial carbon dioxide (P_{aCO_2}) of 68 mm Hg and pH of 7.18 in one study.²⁵ On occasion, the P_{aCO_2} may exceed 100 mm Hg. Hypercapnia despite normal minute ventilation is a result of markedly increased physiologic dead space from alveolar overdistension. The impact of an increase in minute ventilation to lower P_{aCO_2} is mitigated by further increase in dead space related to progressive hyperinflation. As such, hypercapnia in fulminant asthma may not be truly “permissive,” because it may be impossible to normalize P_{aCO_2} through ventilator manipulation.

Given the potential risks of excessive hyperinflation, a reasonable strategy may be to select ventilator settings that typically provide a safe level of dynamic hyperinflation (tidal volume of 7 to 9 mL/kg, respiratory rate of 10 to 14 breaths/min) and accept the resulting P_{aCO_2} (see Table 30-2). Fortunately, hypercapnia in status asthmaticus is usually well tolerated, with serious adverse consequences being uncommon.^{18,19} The most serious consequence of hypercapnia is an increased intracranial pressure in patients with acute neurologic pathology. Although rare, cerebral edema and subarachnoid hemorrhage attributed to hypercapnia has been reported in status asthmaticus.^{41–43} Hypercapnia is of greatest concern when there has been cerebral anoxia secondary to respiratory arrest before intubation. Unfortunately, it may be difficult to achieve normocapnia in fulminant asthma without extracorporeal CO₂ removal unless the patient has rapidly reversible bronchospasm.

One approach to managing acute respiratory acidosis in status asthmaticus is to accept the elevated P_{aCO_2} and administer buffering agents. Unfortunately, standard buffer therapy is relatively inefficient in acute respiratory acidosis. Even partial correction of severe respiratory

acidosis will typically require a minimum of several hundred milliequivalents of sodium bicarbonate in an adult.¹⁸ An animal study did find that administration of sufficient sodium bicarbonate (14 mEq/kg) to preserve a normal pH during induction of acute hypercapnia ($\text{PaCO}_2 = 80$ mm Hg) prevented an increase in cerebral blood flow and intracranial pressure.⁴⁴ The effect on intracranial pressure is less certain if a smaller amount of bicarbonate is given slowly to raise the pH to 7.15 to 7.20, as has been recommended.^{18,44} One problem with giving large amounts of bicarbonate is that once airflow obstruction and hypercapnia resolve, the patient will be left with a therapy-induced metabolic alkalosis.

As a general rule, unless there is some compelling reason to correct underlying respiratory acidosis (e.g., arrhythmia, hyperkalemia, or otherwise unexplained hemodynamic instability), it is probably reasonable to forego attempts to correct serum pH and wait for the PaCO_2 to decrease as airflow obstruction improves. Fortunately, many patients experience substantial improvement in their hypercapnia during the first 12 hours of intubation.⁴⁵ If bicarbonate therapy is given, ideally it should be administered by slow infusion rather than by rapid bolus administration because the latter may lead to an acute increase in CO_2 production and a transient fall in intracellular pH as a consequence of rapid diffusion of CO_2 into cells. An alternative to sodium bicarbonate is **tromethamine** (THAM), a buffer that does not generate CO_2 during the buffering process.¹⁸ Even though THAM may offer some theoretical advantages over sodium bicarbonate for buffering respiratory acidosis, its use may lead to the same problem of a posthypercapnic metabolic alkalosis.

Nonventilator Management

All ventilated patients with severe asthma require inhaled bronchodilators, corticosteroids, and sedation. In rare instances, one or more unconventional approaches also may be considered.

Standard Therapy

Bronchodilators and Glucocorticoids

The optimal dose of inhaled bronchodilators during mechanical ventilation of patients with severe asthma has not been defined. Based on studies of patients with COPD, the optimal dose of albuterol given by metered-dose inhaler is likely to be 4 to 6 puffs.^{46–48} Hourly dosing is appropriate initially. Assessment of lung mechanics during incremental dosing of inhaled β_2 agonists may be useful to determine both the optimal number of puffs and dosing interval for individual patients, using an “ $n = 1$ ” trial. **Ipratropium** might also be considered, given its benefit when added to albuterol therapy in nonintubated asthmatics with severe airflow obstruction.⁴⁹

Glucocorticoids are an essential component of the treatment of severe asthma, with an effect being evident within 12 hours of administration.⁵⁰ An initial dose of 2 mg/kg/day of methylprednisolone or equivalent seems appropriate.⁵¹

Sedation and Paralysis

As with other causes of respiratory failure, minimal goals of sedation in status asthmaticus include provision of anxiolysis, analgesia, and prevention of patient–ventilator dyssynchrony. Patients with status asthmaticus present an additional challenge because of the need to enforce controlled hypoventilation despite acute respiratory acidosis that increases respiratory drive.⁸ A combination of propofol or a benzodiazepine (midazolam or lorazepam) with a narcotic (fentanyl or **morphine**) often proves optimal, and very large doses may be required (see [Table 30-2](#)).⁸ Because airflow obstruction often improves within 24 to 48 hours,^{27,45} residual sedation that delays extubation is undesirable. A major advantage of propofol over benzodiazepines is that propofol allows deep sedation that reverses promptly on its discontinuation. The maximal propofol dose should not exceed 5 mg/kg/hour, because prolonged infusion of very high doses of propofol can lead to life-threatening complications.⁵²

Large doses of sedatives and narcotics in combination with marked lung hyperinflation may result in hypotension. When additional muscle relaxation is needed in the face of hemodynamic instability, it may be safer to administer a nondepolarizing neuromuscular blocking agent than increase the dose of sedatives and narcotics. Even when hypotension is not present, intermittent administration of one or more boluses of a neuromuscular blocking agent may help to provide a period of temporary muscle relaxation, during which time sedative doses can be escalated gradually to achieve target levels. Prolonged use of a neuromuscular blocking agent may increase the likelihood of myopathy in

status asthmaticus (see the section [Death and Complications](#)), but short-term use likely does not carry significant risk.^{53,54} The specific neuromuscular blocking agent selected is likely unimportant, although vecuronium, perhaps, should be avoided in patients with renal failure.⁵⁵ Intermittent bolusing is preferred to a continuous infusion because this permits ongoing assessment of the adequacy of sedation and lessens the likelihood that the patient will undergo unnecessarily prolonged neuromuscular paralysis. When sedatives and narcotics are used liberally, supplemented by short-term intermittent boluses of a neuromuscular blocking agent if needed, very few patients with status asthmaticus will require prolonged, continuous neuromuscular paralysis.

Alternative Therapies

Occasionally, one of several nontraditional approaches may be considered in patients with fulminant asthma. Strategies that have been reported anecdotally to be beneficial include the use of heliox, inhalational anesthetics, or ketamine, [nitric oxide](#), mucolytic agents, bronchoscopy, and extracorporeal support.

Heliox

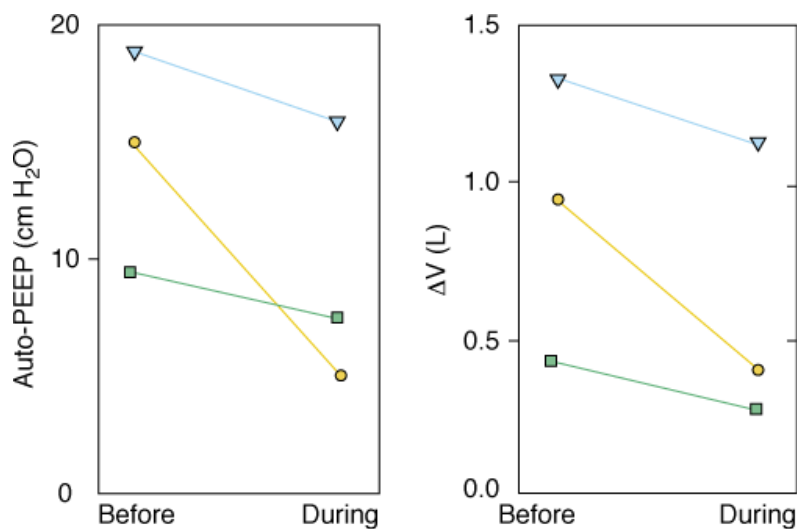
The lower gas density of heliox reduces frictional resistance where gas flow is turbulent and, by lowering the Reynolds number, also encourages laminar flow.¹¹ One study reported that heliox produced a rapid fall in peak airway pressure and PaCO_2 of ventilated patients with asthma.⁵⁶ The effects on P_{PLAT} and auto-PEEP, however, were not reported, and it is unclear whether changes in PaCO_2 were at constant minute ventilation.⁵⁶ A second study of heliox in severe asthma found little change in PaCO_2 when tidal volume and respiratory rate were held constant.⁵⁷ Two prospective studies found that a 70:30 mixture of heliox reduced auto-PEEP during mechanical ventilation of patients with COPD.^{58,59} This suggests that heliox may have a measurable benefit in selected patients and might be considered for use when conventional management results in an unacceptable degree of dynamic hyperinflation in patients with status asthmaticus.²³ Continuous use of heliox is very expensive and can only be justified if it significantly improves indices of hyperinflation (P_{PLAT} , auto-PEEP) or hypercapnia.⁶⁰

Before using heliox, it is crucial to fully understand how its use will affect performance of the ventilator.⁶¹ Use of a density-independent spirometer in the expiratory limb of the ventilator circuit is advisable to ensure accurate setting of tidal volume.¹¹

General Anesthetics

Inhalational anesthetics have potent bronchodilating properties, and several anecdotal reports have described their use in status asthmaticus.^{62,63} In the only study to carefully assess lung mechanics, isoflurane resulted in a decrease in airways resistance and auto-PEEP in three patients with asthma, although only one had a marked response ([Fig. 30-10](#)).⁶⁴

Figure 30-10



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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The effect of isoflurane on auto-PEEP and change in lung volume (ΔV) in three patients with severe asthma. (Adapted, with permission, from Maltais F, Sovilj M, Goldberg P, Gottfried SB. Respiratory mechanics in status asthmaticus. Effects of inhalational anesthesia. *Chest*. 1994;105:1401–1406.)

Isoflurane and sevoflurane are less arrhythmogenic than halothane and have equal or greater bronchodilator properties.⁶³ All these agents may cause hypotension secondary to their peripheral vascular effects, and an increase in venous capacitance may be particularly detrimental when marked dynamic hyperinflation already has compromised venous return. The adverse hemodynamic effects generally can be mitigated through liberal administration of fluid and with vasoactive support if necessary. It is mandatory, of course, that personnel highly skilled in the use of anesthetic agents be responsible for their administration. Some intensive care unit ventilators have a port to which the anesthesia vaporizer can be attached, and with appropriate scavenging of anesthetic gases, the general anesthetic can be administered in the intensive care unit.⁶²

Ketamine, an intravenous dissociative anesthetic, also has been advocated for use in severe asthma. This drug, however, can lead to significant increases in blood pressure, heart rate, and intracranial pressure. There seems little justification for the use of ketamine in ventilated patients with status asthmaticus.

Mucolytics

The potential benefit of *N*-acetylcysteine as a mucolytic is unknown; in an anecdotal report, it seemed to enhance bronchoscopic extraction of mucous plugs.⁶⁵ Benefit from rhDNAse also has been reported.⁶⁶

Bronchoscopy

Patients with fatal asthma often have extensive mucoid impaction.⁶⁷ Removal of impacted mucus by bronchoscopy has been reported to lower airway pressures and improve gas exchange in ventilated patients with severe asthma.⁶⁵ However, there is a potential for worsening bronchospasm and several large series have reported good outcomes without use of bronchoscopy.^{18,24,26,32,45} Patients who fail to improve after several days of mechanical ventilation might be considered for diagnostic bronchoscopy to inspect the airways for mucous plugs that might be extracted, the goal being to reduce the duration of ventilator support.

Extracorporeal Life Support

Both pump-driven and pump-less extracorporeal life support has been used in severe asthma.^{68–70} Because severe asthma is fully reversible, this approach clearly would be justified if there were an imminently lethal impairment in gas exchange. Refractory hypoxemia, however, is unusual in asthma and hypercapnia generally is well tolerated (see the section [Gas Exchange: Hypercapnia](#), above). Extracorporeal life support might be considered if there is profound hypercapnia and extreme hyperinflation, especially if accompanied by

barotrauma or hemodynamic instability.^{69,70} Besides correcting hypercapnia and avoiding complications related to hyperinflation, extracorporeal life support could also permit safe use of bronchoscopy to treat mucoid impaction.

Death and Complications

Mortality

Published mortality rates for patients undergoing mechanical ventilation for severe asthma have varied greatly, ranging from 0% to 38%.^{32,45,71–74} A literature review of more than 1220 patients reported an average mortality of 12.4%.²³ Although the outcome seems to have improved over the last two decades, perhaps because of more widespread use of controlled hypoventilation, mortality rates as high as 15% to 20% have been reported in series published during the last decade.^{72,73} Most fatalities result from cerebral anoxia secondary to cardiorespiratory arrest before intubation. Indeed, in a recent analysis of 1223 patients who underwent mechanical ventilation for severe asthma, 80% of in-hospital deaths were preceded by cardiorespiratory arrest before admission to the intensive care unit.⁷³

Complications

Patients with severe asthma are at risk for many of the same complications affecting other ventilated patients, including atelectasis, nosocomial pneumonia, sinusitis, pulmonary embolism, and gastrointestinal bleeding. Additional complications may result directly from the asthma exacerbation or from medications used to treat airflow obstruction and provide muscle relaxation; these include ventilator-associated hypotension, barotrauma, myocardial injury, rhabdomyolysis, lactic acidosis, neurologic injury, and acute myopathy (Table 30-4).

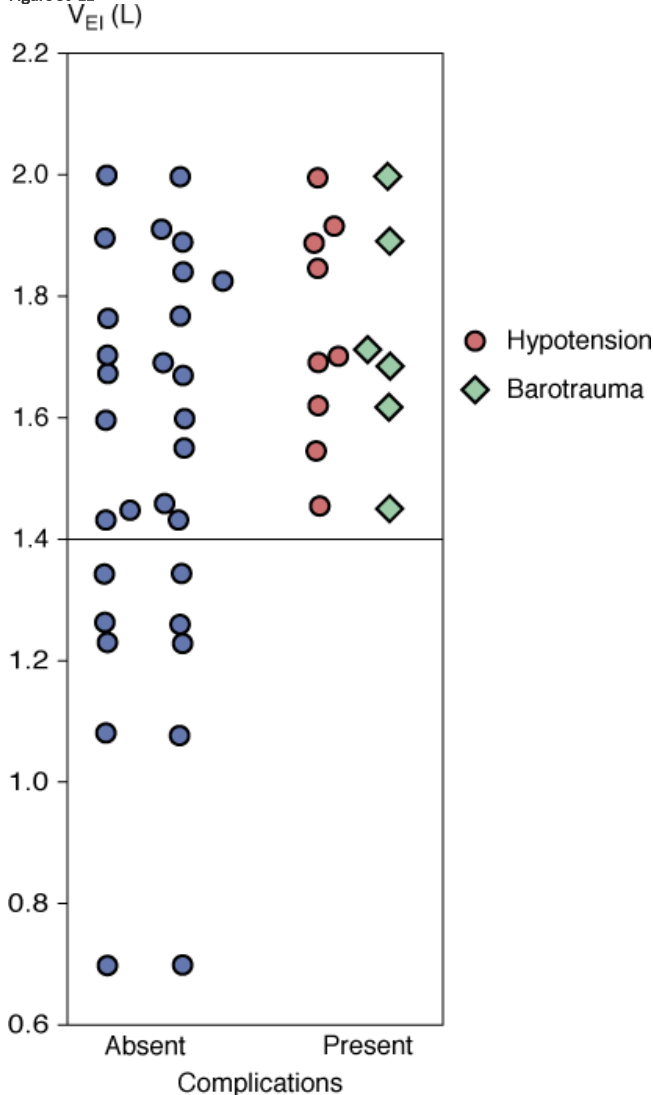
Table 30-4: Complications of Mechanical Ventilation for Severe Asthma

Complications	Likely Mechanism
Hypotension	Primary: excessive hyperinflation, sedatives
	Secondary: pneumothorax, myocardial depression
Barotrauma	Excessive hyperinflation
Myocardial dysfunction	Primary: stress cardiomyopathy secondary to massive catecholamine release
	Secondary: severe myocardial hypoxia and/or acidosis
Rhabdomyolysis	Primary: extreme muscle exertion with or without hypoxia
	Secondary: high-dose propofol
Lactic acidosis	Primary: excessive β_2 agonists
	Secondary: extreme muscle exertion and/or hypoxia
CNS injury	Primary: cerebral anoxia secondary to respiratory arrest
	Secondary: hypercapnia-related cerebral edema, subarachnoid hemorrhage
Acute myopathy	Glucocorticoids plus prolonged paralysis or deep sedation

Hypotension

Hypotension during mechanical ventilation for status asthmaticus is most often a result of the combined effects of sedatives and the excessive pulmonary hyperinflation that impedes venous return. In one series, mild-to-moderate hypotension was documented at some point during the course of mechanical ventilation in 35% of patients, with risk being greatest in patients whose V_{EI} exceeded approximately 20 mL/kg (Fig. 30-11).²⁴ Extreme hyperinflation can lead to cardiac arrest with pulseless electrical activity.¹⁷ When a ventilated patient with severe asthma develops significant hypotension, a 1-minute apnea trial is recommended.²³ If the apnea trial and a rapid infusion of fluid do not restore blood pressure, then less-common causes of hypotension (e.g., pneumothorax, myocardial depression) must be considered.

Figure 30-11



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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Relationship between hypotension, barotrauma, and lung volume at end inspiration (V_{EI}). (Used, with permission, from Williams TJ, Tuxen DV, Scheinkestel CD, et al. Risk factors for morbidity in mechanically ventilated patients with acute severe asthma. *Am Rev Respir Dis*. 1992;146:607–615.)

Barotrauma

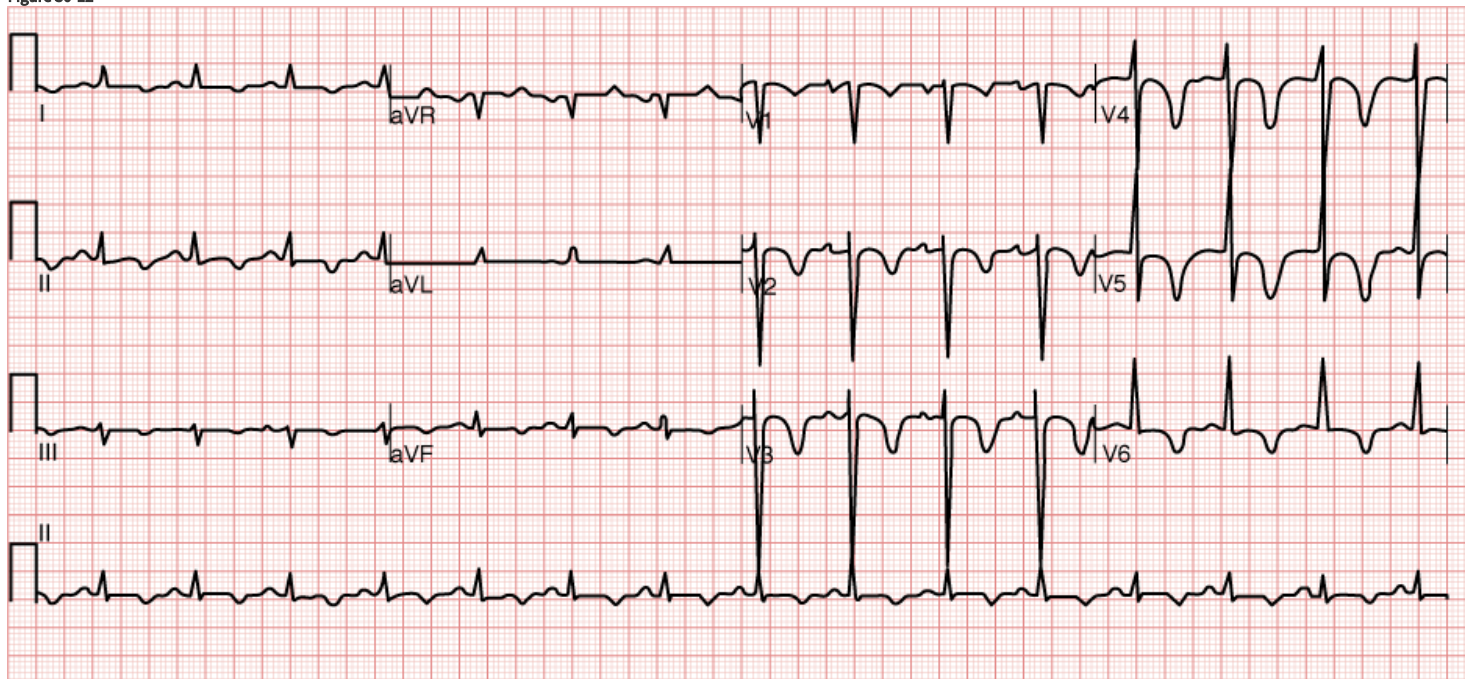
Pneumothorax also occurs most often in patients with the highest end-inspiratory lung volume (see Fig. 30-11).²⁴ The incidence of pneumothorax was as high as 30% in some early series,⁷¹ but is relatively infrequent (<10%) when a strategy of controlled hypoventilation is used.^{8,26,27,32,45,74} In an analysis of barotrauma in various types of respiratory failure, pneumothorax was documented in only 6% of patient with status asthmaticus.⁷⁵ A pneumothorax may be particularly dangerous in patients with severe asthma because the hyperinflated lungs resist collapse and allow even a small pneumothorax to be under tension, resulting in rapid deterioration and sometimes death.²³ For

this reason, clinical (and radiographic) diagnosis of tension pneumothorax may be challenging.²³ Chest tubes always should be placed by blunt dissection rather than by the blind trocar method to avoid injuring the hyperinflated lungs.²³

Cardiac Complications

A reported cardiac complication of status asthmaticus is stress cardiomyopathy, typically manifested as decreased myocardial contractility with reversible segmental myocardial wall-motion abnormalities and deep T-wave inversions in the electrocardiogram, simulating myocardial ischemia (Fig. 30-12).^{76,77} Stress cardiomyopathy in status asthmaticus is similar to that seen after subarachnoid hemorrhage, severe psychological stress, and other conditions associated with massive endogenous sympathetic activation. It often follows a benign course with no long-term cardiac sequelae, but may be associated with significant hypotension.⁷⁷

Figure 30-12



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Electrocardiogram of a patient with acute reversible left-ventricular dysfunction caused by status asthmaticus. Deep inverted T-waves in anterior precordial leads are seen commonly in this condition.

Rhabdomyolysis

Rhabdomyolysis has been reported in patients with status asthmaticus, presumably as a result of extreme muscular exertion coupled with hypoxia.⁷⁸ Rhabdomyolysis also has been noted in patients who had received prolonged infusions of propofol in very high doses.⁵²

Lactic Acidosis

Mild lactic acidosis is relatively common in status asthmaticus and has been attributed to lactate production by respiratory muscles.⁷⁹⁻⁸¹ Lactic acidosis can also occur as a result of excessive use of β_2 agonists. Although more common when the latter are given intravenously, lactic acidosis also can follow high-dose inhalational therapy with albuterol.⁸² Even a moderate degree of lactic acidosis may be problematic in patients with significant hypercapnia.

Central Nervous System Injury

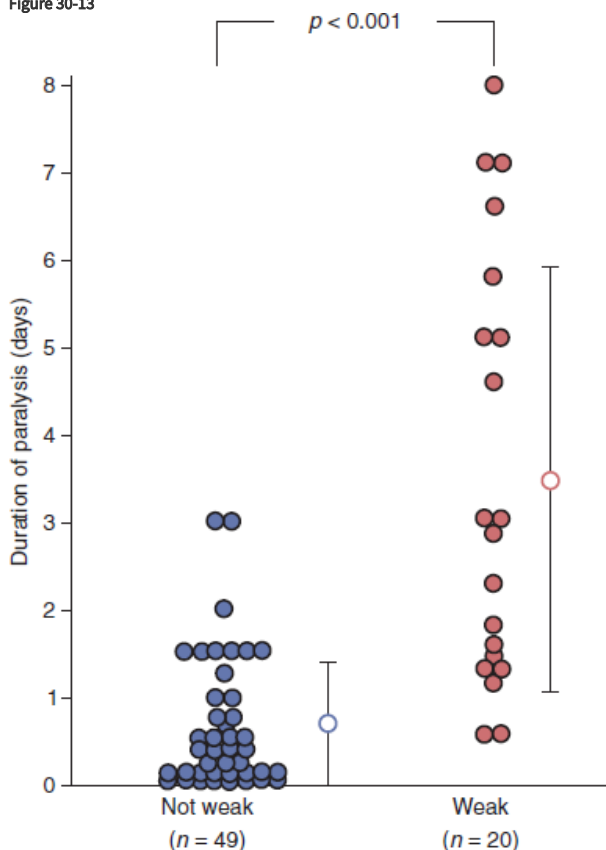
As noted earlier, cerebral anoxia is the most common cause of death in patients with status asthmaticus.⁷³ Rarely, cerebral edema or subarachnoid hemorrhage in status asthmaticus can occur in the absence of prior cerebral anoxia and have been attributed to

hypercapnia.^{41–43}

Myopathy

Acute myopathy probably is the most common cause of morbidity affecting patients with asthma who undergo mechanical ventilation. The pathogenesis of myopathy is incompletely understood, but has been attributed to the combined effects of glucocorticoids and prolonged neuromuscular paralysis (Fig. 30-13).^{53,54} Myopathy also occurs in patients with asthma who have undergone prolonged (>5 to 7 days) mechanical ventilation under deep sedation without paralysis.^{83,84} It is possible that prolonged near-total muscle inactivity, whether induced by neuromuscular paralysis or by deep sedation, may increase the risk of this complication.⁸⁴ Recent studies show that daily physical therapy during scheduled awakening significantly reduces the incidence of intensive care unit-acquired weakness.⁸⁵ When patients with asthma are unable to be extubated within a few days of intubation, daily scheduled awakening for physical therapy should be strongly considered, provided airflow obstruction is not so severe that withdrawal of sedation is deemed unsafe.

Figure 30-13



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Duration of paralysis in patients with and without weakness (defined clinically) after undergoing mechanical ventilation for severe asthma. (Used, with permission, from Leatherman JW, Fluegel WL, David WS, et al. Muscle weakness in mechanically ventilated patients with severe asthma. *Am J Respir Crit Care Med*. 1996;153:1686–1690.)

Posthospitalization Prognosis

Although in-hospital mortality of patients who receive mechanical ventilation for severe asthma is relatively low, one study reported that seventeen of 121 (14%) patients died of a recurrent asthma attack during a follow-up period of 6 years, with most deaths occurring in the first year after hospital discharge.⁸⁶ Another study identified prior intubation as the strongest risk factor for death from asthma.¹ These data emphasize the crucial importance of outpatient management—including regular use of inhaled glucocorticoids, avoidance of smoking and other factors known to increase airway responsiveness, close supervision, and intensive education—following hospital discharge of patients with asthma who undergo mechanical ventilation.

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